

Linearized fluxes inside shell sector for 1D, 2D and 3D droplets. We solve the Diffusion equation with a constant source in the shell sector and check for the passive case too.

```
In[ ]:= $Assumptions = Diff > 0 && cEqOut > 0 && R > 0 && L > 0
```

```
Out[ ]:= Diff > 0 && cEqOut > 0 && R > 0 && L > 0
```

Solve the stationary state equation for 1D droplets

```
In[ ]:= sol1D[x_] = FullSimplify[
    c[x] /. First@DSolve[{0 == Diff * Laplacian[c[x], {x}, "Cartesian"] + A,
        c[R] == cEqOut, c[R + L] == cfar}, c, x]]
Out[ ]:= 
$$\frac{-((2 \text{cfar Diff} + A L (L + R - x)) (R - x)) + 2 \text{cEqOut Diff} (L + R - x)}{2 \text{Diff} L}$$

```

Check the boundary conditions

```
In[ ]:= FullSimplify@sol1D[R]
```

```
Out[ ]:= cEqOut
```

```
In[ ]:= FullSimplify@sol1D[R + L]
```

```
Out[ ]:= cfar
```

Check the solution by plugging it in the equation

```
In[ ]:= FullSimplify[Diff * Laplacian[sol1D[x], {x}, "Cartesian"] + A]
```

```
Out[ ]:= 0
```

Calculate line integrated fluxes at the droplet surface

```
In[*]:= FluxInBound1D = FullSimplify[FullSimplify[-2 * Diff * D[sol1D[x], x] /. x -> R]]
Out[*]:= 
$$\frac{2 (cEqOut - cfar) Diff}{L} - A L$$

```

```
In[*]:= FullSimplify[Limit[FluxInBound1D, A -> 0]]
Out[*]:= 
$$\frac{2 (cEqOut - cfar) Diff}{L}$$

```

Solve the stationary state equation for 2D droplets

```
In[*]:= sol2D[r_] =
  FullSimplify[c[r] /. First@DSolve[{0 == Diff * Laplacian[c[r], {r, ϕ}, "Polar"] + A,
    c[R] == cEqOut, c[R + L] == cfar}, c, r]]
Out[*]:= 
$$\frac{1}{4 Diff \log\left[\frac{R}{L+R}\right]} \left( (4 cEqOut Diff - 4 cfar Diff - A L (L + 2 R)) \log[r] + \right.$$


$$\left. (4 cfar Diff + A (L - r + R) (L + r + R)) \log[R] + \right.$$


$$\left. (-4 cEqOut Diff + A (r - R) (r + R)) \log[L + R] \right)$$

```

Check the boundary conditions

```
In[*]:= FullSimplify@sol2D[R]
Out[*]:= cEqOut
```

```
In[*]:= FullSimplify@sol2D[R + L]
Out[*]:= cfar
```

Check the solution by plugging it in the equation

```
In[*]:= FullSimplify[Diff * Laplacian[sol2D[r], {r, ϕ}, "Polar"] + A]
Out[*]:= 0
```

Calculate perimeter integrated fluxes at the droplet surface

```
In[*]:= FluxInBound2D = FullSimplify[FullSimplify[-2 π * R * Diff * D[sol2D[r], r] /. r -> R]]
Out[*]:= 
$$A \pi R^2 + \frac{\pi (-4 cEqOut Diff + 4 cfar Diff + A L (L + 2 R))}{2 \log\left[\frac{R}{L+R}\right]}$$

```

In[]:= FortranForm[FluxInBound2D]

Out[]:=FortranForm=

$$\begin{aligned} & A \pi R^2 + \\ & - \left(\pi \left(-4 c_{\text{EqOut}} \text{Diff} + 4 c_{\text{far}} \text{Diff} + \right. \right. \\ & - \left. \left. A L (L + 2R) \right) \right) / \\ & - \left(2 \log(R / (L + R)) \right) \end{aligned}$$

In[]:= FullSimplify[Limit[FluxInBound2D, A → 0]]

Out[]:=
$$\frac{2 (-c_{\text{EqOut}} + c_{\text{far}}) \text{Diff} \pi}{\log\left[\frac{R}{L+R}\right]}$$

Solve the stationary state equation for 3D droplets

In[]:= sol3D[r_] = FullSimplify[

c[r] /. First@DSolve[{0 == Diff * Laplacian[c[r], {r, ϕ, θ}, "Spherical"] + A,
c[R] == cEqOut, c[R + L] == cfar}, c, r]]

Out[]:=
$$\frac{1}{6 \text{Diff} L r} \left(6 c_{\text{far}} \text{Diff} (r - R) (L + R) + (L - r + R) (6 c_{\text{EqOut}} \text{Diff} R + A L (r - R) (L + r + 2 R)) \right)$$

Check the boundary conditions

In[]:= FullSimplify@sol3D[R]

Out[]:= cEqOut

In[]:= FullSimplify@sol3D[R + L]

Out[]:= cfar

Check the solution by plugging it in the equation

In[]:= FullSimplify[Diff * Laplacian[sol3D[r], {r, ϕ, θ}, "Spherical"] + A]

Out[]:= 0

Calculate surface area integrated fluxes at the droplet surface

In[]:= FluxInBound3D =

FullSimplify[FullSimplify[-4 π * R^2 * Diff * D[sol3D[r], r] /. r → R]]

Out[]:=
$$-\frac{2 \pi R \left(-6 c_{\text{EqOut}} \text{Diff} (L + R) + 6 c_{\text{far}} \text{Diff} (L + R) + A L^2 (L + 3 R) \right)}{3 L}$$

In[]:= FortranForm[FluxInBound3D]

Out[]:=FortranForm=

$$\begin{aligned} & (-2 \pi R (-6 c_{\text{EqOut}} \text{Diff} (L + R) + \\ & - 6 c_{\text{far}} \text{Diff} (L + R) + \\ & - A L^2 (L + 3 R)) / (3 L) \end{aligned}$$

```
In[*]:= FullSimplify[Limit[FluxInBound3D, A → 0]]
```

```
Out[*]= 
$$\frac{4 (cEqOut - cfar) \text{Diff} \pi R (L + R)}{L}$$

```
